

OOP Exercise

Project #1A

Create a class book with the following attributes

- i. ID
- ii. Title
- iii. Price
- iv. Type - any one of PRINT, E-book
- v. Authors - a book can be written by more than one authors
- vi. Publisher
- vii. Publish Date
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- Setting author and getting author list must be through an interface

An ITBook inherits from Book with the following attributes

- i. Brief Description
- ii. Tags - a book can be associated with multiple tags
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- Setting tags and getting tag list must be through an interface
- The book must implement IComparable<T> interface so that it compared by title or price

Project requirements:

- Use of abstract class, overrides
- Use of Enumeration
- Constructor overloading
- Interface implementations
- Creation of instances and storing arrays
- Sorting array items

- LINQ queries: Ordering, Filtering and grouping [Use both query expressions and extension methods]

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Project #1B

Create a class book with the following attributes

i. ID

ii. Title

iii. Price

iv. Type - any one of PRINT, EBook

v. Authors - a book can be written by more than one authors

vi. Publisher

vii. Publish Date

viii. a book can be associated with multiple tags

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An ITBook inherits from Book with the following attributes

i. Tags - a book can be associated with multiple tags

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Create a Generic interface to get author names in a string separated by comma and Implement the

interface in IT Book

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Create a Generic interface to get tags in a string separated by comma and Implement the interface in IT

Book

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Project #1C

- Create a IT Book collection implementing IEnumerable

- You must reuse classes from project #1B (adding reference)

Project requirements:

- Use of generic constraints
- Creation of instances
- Use of related classes
- Use of yield in Enumerator
- Testing created collection